

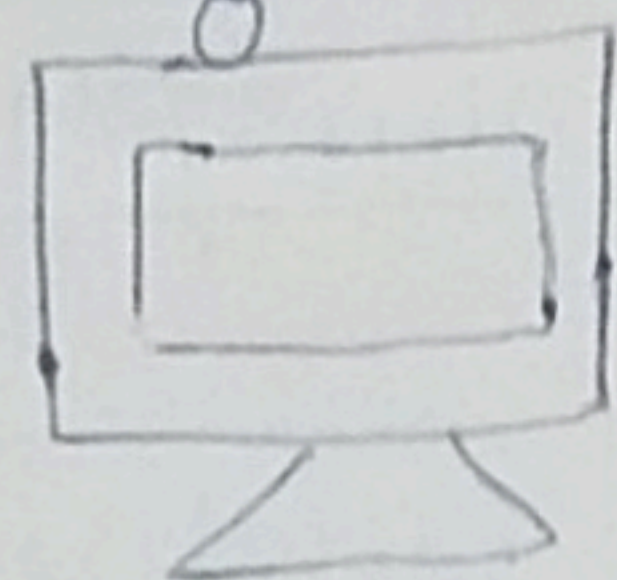
Chapter - 9

File I/O

The random-access memory is volatile, and all its contents are lost once a program terminates in order to persist the data forever, we use files.

A file is data stored in a storage device. A Python program can talk to the file by reading content from it and writing content to it.

Programmer



Computer program
written in Python

Write

Read

File

RAM = Volatile

HDD = Non-Volatile

● Types of Files

1. Text files (.txt, .c etc)
2. Binary files (.jpg, .dat, etc)

Python has a lot of functions for reading, updating, and deleting files.

File.txt

f = open('File.txt')

data = f.read()

print(data)

f.close()

file created in vs code

open with python in vs code

● Opening a File

Python has an `open()` function for opening files. It takes 2 parameters: filename and mode.

```
#open("filename", "mode of opening (read mode by default)")
```

```
open("this.txt", "r")
```

```
# Read its contents
```

```
text = f.read()
```

```
print(text)
```

```
# print its contents
```

```
f.close()
```

```
# close the file
```

> We can also use `f.readline()` function to read one full line at a time.

```
f.readline()
```

● Modes of Opening a File

r - opening for reading

w - open for writing

a - open for appending

+ - open for updating

'rb' - will open for read in binary mode

'rt' - will open for read in text mode.

● Write Files in Python

In order to write a file, we first open it in write or append mode after which, we use the python's `f.write()` method to write to the file!

open the file in write mode

```
f = open("this.txt", "w")
```

write a string to the file

```
f.write("this is nice")
```

close the file

```
f.close()
```

● Examples of Readlines and Readline Method in File

● Readlines Method

```
f = open("file.txt")
```

```
lines = f.readlines()
```

```
print(lines, type(lines))
```

```
f.close()
```

types will give us list.

● Readline Method

```
f = open("file.txt")
```

```
line1 = f.readline()
```

```
print(line1, type(line1))
```

```
line2 = f.readline()
```

```
print(line2, type(line2))
```

```
f.close()
```

types will give us string

• With Statement

The best way to open and close the file automatically is the with statement.

Open the file in read mode using 'with', which automatically closes the file

```
with open("this.txt", "r") as f:
```

```
    # Read the content of the file
```

```
    text = f.read()
```

```
    # print the contents
```

```
    print(text)
```

Chapter 9

Practice Set

- 1) Write a program to read the text from a given file "poem.txt" and find out whether it contains the word "mink".
- 2) The game() function in a program let's user play a game and returns the score as an integer. You need a read file "Hi-score.txt" which is either blank or contains the previous Hi-score. You need to write a program to update the Hi-score whenever the game() function breaks the Hi-score.
- 3) Write a program to generate multiplication table from 2 to 20 and write it to the different files. Place these files in a folder for a 13-year old.
- 4) A file contains a word "Donkey" multiple times. You need to write a program which replaces this word with ##### by updating the same file.
- 5) Repeat a program 4 for a list of such words to be censored.
- 6) Write a program to mine a log file and find out whether it contains "python".
- 7) Write a program to find out the line numbers where Python is presented from Ques 6.

- 8) Write a program to make a copy of a text file "this.txt".
- 9) Write a program to find out whether a file is identical & matches the content of another file.
- 10) Write a program to write out the content of a file using Python.
- 11) Write a python program to rename a file to "renamed-by-python.txt".